ECE 340 # Homework Assignment #6 **Fall 2026**

DUE: FRIDAY, OCTOBER 16TH, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME AT 5:00PM ON THE DUE DATE.

1. STEADY-STATE MINORITY-CARRIER DIFFUSION

Consider a long, uniformly doped n-type silicon sample at $T = 300$ K. The sample occupies $x \geq 0$ and may be treated as semi-infinite. The material parameters are

$$N_D = 1.0 \times 10^{16}\ \text{cm}^{-3}, \qquad n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}, \qquad \mu_p = 450\ \text{cm}^2/\text{V}\cdot\text{s},$$

and the minority-hole lifetime is $\tau_p = 2.0\ \mu$s. At $x = 0$, illumination maintains an excess hole concentration $\delta p(0) = 1.0 \times 10^{14}\ \text{cm}^{-3}$. For $x > 0$, there is no optical generation and no applied electric field. Assume steady-state conditions, low-level injection, and $\delta p(x) \to 0$ as $x \to \infty$. Use $kT/q = 0.0259$ V and $q = 1.60 \times 10^{-19}$ C.

(A). Calculate the equilibrium minority-hole concentration p_{n0}. Verify that the specified excess carrier concentration satisfies the low-level-injection condition.

(B). Use the Einstein relation to calculate the hole diffusion coefficient D_p and the hole diffusion length L_p.

(C). Starting with the steady-state minority-carrier diffusion equation

$$D_p \frac{d^2 \delta p}{dx^2} - \frac{\delta p}{\tau_p} = 0,$$

derive $\delta p(x)$. At what distance from $x = 0$ has the excess hole concentration fallen to 10% of its value at $x = 0$?

(D). Determine the hole diffusion current density $J_p(x)$. Calculate J_p at $x = 0$ and at $x = L_p$, and state the direction of the conventional hole current.

(E). Show that the total recombination current per unit area in the region $x > 0$,

$$J_{\text{rec}} = q \int_0^\infty \frac{\delta p(x)}{\tau_p}\, dx,$$

equals the hole diffusion current entering the region at $x = 0$. Briefly explain the physical meaning of this result.

2. P-N JUNCTIONS IN EQUILIBRIUM

Consider a GaAs p-n diode that is asymmetrically doped with $N_D = 4 \times 10^{16} cm^{-3}$ and $N_A = 1.6 \times 10^{17} cm^{-3}$. Answer the following questions assuming the diode is in equilibrium at $T = 300K$. For GaAs, $E_g = 1.43eV$, $n_i = 2.1 \times 10^6 cm^{-3}$, and $\epsilon = 12.9\epsilon_0$.

(A). What is the equilibrium condition as it pertains to both the Fermi level and net current flow?

(B). Calculate the contact potential V_0.

(C). Calculate the depletion width in the n-region (x_{n0}), the depletion width in the p-region (x_{p0}), and the overall depletion width (W).

(D). Calculate the electrostatic potential **relative to the p-side** at the metallurgic junction $V(x = 0)$.

(E). Plot the electric field $E(x)$ as a function of x and calculate the peak value.

(F). Plot the band diagram paying attention to the shape of the bands in the junction and labeling E_i, E_C, E_V, and E_F.

3. JUNCTION PROPERTIES

Consider two p^+n junctions. One has an n-doping of N_D and the other has an n-doping of $\frac{1}{2}N_D$. Everything else remains unchanged. How do the following quantities change in the second junction compared to the first one? Indicate only increase or decrease and explain your answer for each **mathmatically**.

(A). Total amount of depletion charge.

(B). Magnitude of the peak electric field.

(C). Depletion width on the n-side (x_n).

(D). Depletion width on the p-side (x_p).

(E). Built-in potential (V_0).

4. Si n–p Junction in Equilibrium

An *abrupt* silicon **n–p** (note the order) junction has a circular cross–section of diameter $10\,\mu$m. The doping and material parameters are:

$$N_D = 4.0\times10^{16}\ \text{cm}^{-3}, \qquad N_A = 1.5\times10^{17}\ \text{cm}^{-3}, \qquad n_i = 1.5\times10^{10}\ \text{cm}^{-3},$$

and the silicon permittivity is $\varepsilon_s = \varepsilon_{\text{Si}} = 11.8\,\varepsilon_0$. Assume $T = 300\,\text{K}$ unless otherwise specified.

(A). Calculate the Fermi–level positions in each quasi–neutral region at equilibrium: $(E_F - E_i)$ in the n region and $(E_i - E_F)$ in the p region. Then compute the contact (built-in) potential V_0 of the junction.

(B). Sketch the equilibrium energy-band diagram of the **n–p** junction (conduction band, valence band, intrinsic level, Fermi level, and built-in potential). Clearly label E_C, E_V, E_i, E_F, and qV_0.

(C). Determine the depletion width W, the depletion extents x_{n0} and x_{p0} on the n and p sides, respectively, the total uncovered ionized charge Q_+ in the depletion region, and the maximum electric field magnitude E_0. Use $\varepsilon_s = 11.8\,\varepsilon_0$ and take q as the elementary charge.